



A decision support model for nursing chair design driven by patent literature analysis and trapezoidal fuzzy AHP

Jun Du^a, Jiangang Zhu^{a,b,*} 

^a College of Furnishings & Industrial Design, Nanjing Forestry University, Nanjing, China

^b Jiangsu Co-innovation Centre of Efficient Processing and Utilization of Forest Resources, Nanjing, China

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ABSTRACT

The postpartum period presents significant ergonomic challenges for women. Prolonged and frequent breastfeeding often leads to fatigue and musculoskeletal disorders (MSDs), affecting maternal and infant health. Our preliminary study shows that 78 % of breastfeeding mothers experience lower back discomfort after just 15 min, and 62 % suffer from shoulder and neck tension due to forward-leaning postures. Current nursing chair designs struggle to meet postpartum needs due to the lack of standardized evaluation tools and user-centered design approaches. This study aims to address the significant ergonomic challenges faced by postpartum women by developing a novel decision-support model that integrates Patent Literature Analysis (PLA) and the Trapezoidal Fuzzy Analytic Hierarchy Process (AHP). PLA identifies key innovation trends in nursing chair design, while fuzzy AHP ranks evaluation criteria, forming a dynamic framework of “technology trend insights, quantitative indicator evaluation.” The model effectively aligns technical feasibility with user needs, providing a robust framework for optimizing nursing chair designs to reduce musculoskeletal disorders (MSDs) and enhance maternal-infant health. PLA clarifies technical boundaries by capturing evolution trends, while fuzzy AHP quantifies priority levels of user requirements and addresses uncertainties in subjective evaluations using fuzzy numbers. This enables a continuous design loop of “trend updates, weight adjustments, solution optimization.” Evaluation results highlight “structural rationality” (weight 0.2195) and “safety” (weight 0.1272) as top priorities. The optimized design achieved a score of 0.6669, significantly enhancing user comfort and product competitiveness. This framework offers a user-driven, technically grounded path for nursing chair optimization, aiming to reduce MSDs and promote maternal-infant well-being.

1. Introduction

The design optimization of maternal and infant health products must fully account for the dynamic needs of women during their unique physiological stages. Throughout the continuous physiological process from late pregnancy to postpartum lactation, women undergo a series of changes including shifts in body center of gravity, increased ligament laxity, and ongoing musculoskeletal adaptations. These changes present distinctive biomechanical-ergonomic challenges in maternal and newborn care contexts: Sunaga et al. (2020) found that women in late pregnancy exhibit significant postural control alterations during sit-to-stand motions, including reduced center of pressure displacement and modified load distribution. Milligan (1989) found that breastfeeding, previous depressive moods, and having a difficult-to-care-for infant are predictors of fatigue within the first six weeks postpartum.

Breastfeeding, a special activity characterized by long duration, high frequency, and irregular timing, can easily lead to physical and mental fatigue for breastfeeding women. Our preliminary user study ($n = 50$) involved asking participants to simulate their breastfeeding postures while sitting on chairs, aiming to understand users' behavioral habits. Through the recording of users' breastfeeding postures (Fig. 1) and the results of the survey questionnaire, it was found that 78 % of women reported experiencing varying degrees of lumbar discomfort after 15 min of breastfeeding, and 62 % of women stated that they had used forward-leaning breastfeeding postures, which tend to cause shoulder-neck muscle tension or abdominal compression. These physiological adaptations indicate that prolonged sitting in improper positions exacerbates physical strain on perinatal women, potentially leading to excessive lumbar curvature, muscle tension, and even musculoskeletal disorders (MSDs), thereby placing them in a suboptimal

* Corresponding author. College of Furnishings & Industrial Design, Nanjing Forestry University, Nanjing, China.

E-mail address: austin_zhu@njfu.edu.cn (J. Zhu).

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health state.

Selecting an appropriate nursing chair can alleviate discomfort and fatigue during breastfeeding. However, such specialized chairs are not yet widely available on the market, leading many mothers to resort to ordinary household seating for nursing. A cross-sectional study conducted in the Dschang Health District, Cameroon, involving 422 lactating mothers revealed an association between breastfeeding duration and the occurrence of spine musculoskeletal disorders (MSDs): mothers who breastfed for more than 30 min per session had a significantly higher risk of developing spine MSDs compared to those with sessions shorter than 30 min (Womsi et al., 2025). Meanwhile, our survey also found that 78 % of users experienced noticeable lower back discomfort after maintaining the forward-leaning posture commonly adopted during breastfeeding for more than 15 min. As a result, standard chairs fail to mitigate the static muscle load caused by breastfeeding postures and lack the necessary dynamic stability for reclined positions.

Therefore, the motivation of this study is to address the ergonomic shortcomings of current nursing chairs and the absence of standardized evaluation tools. The research aims to establish a decision-support framework that integrates Patent Literature Analysis (PLA) and the Trapezoidal Fuzzy Analytic Hierarchy Process (AHP). PLA identifies emerging technological trends in nursing chair design, while fuzzy AHP evaluates and ranks user-centered criteria, together forming a dynamic model of “technology trend insights – quantitative indicator evaluation.” This framework not only supports manufacturers in balancing technological feasibility with user needs but also contributes to maternal and infant health. By doing so, it directly advances Sustainable Development Goal 3, ensuring healthy lives and promoting well-being for all at all ages, and Sustainable Development Goal 9, fostering industry, innovation and infrastructure, thereby reinforcing the academic, industrial, and societal significance of the research (United Nations, 2023).

2. Literature review

2.1. Musculoskeletal disorders (MSDs) caused by postpartum fatigue

It has been found that the breastfeeding period for infants generally lasts from six months to over a year, with a high frequency of daily feedings. For infants under three months old, it is necessary to feed them 8 to 12 times a day; for infants aged 4–6 months, they should be fed approximately every 4–5 h; and for infants aged 7–9 months, they typically feed about 3–4 times a day. Most breastfeeding sessions last about 15–20 min (Meek et al., 2022). Ojukwu et al. (2021) demonstrated that a high prevalence of musculoskeletal disorders (MSDs) related to breastfeeding has been reported among nursing mothers. Musculoskeletal disorders (MSDs) refer to chronic health issues caused by prolonged exertion during occupational activities (Changlei and Qiang, 2008). The basic factors of chronic musculoskeletal disorders include localized static muscle loading and repetitive motion (Shanshan, 2023). For example, mismatched classroom furniture significantly increases the risk of neck, shoulder, and lower back pain in students (Kahya, 2019), which shares similar biomechanical mechanisms with the musculoskeletal issues experienced by breastfeeding women due to poorly designed seating. Among students who use mismatched furniture

over extended periods, the incidence of MSD symptoms is significantly higher in senior cohorts, demonstrating the cumulative effects of chronic musculoskeletal strain. poor working postures, prolonged fixed positions, mental stress, adverse environments, excessive family burdens, and poor ergonomic design can all exacerbate chronic musculoskeletal injuries (Liu Changjun and Wang, 2004).

It is worth noting that the impact of breastfeeding on mothers is not only reflected at the level of physiological load, but psychological factors also exert significant influences through endocrine regulatory mechanisms (Josefson et al., 2023). Maternal psychological distress such as stress and anxiety can interfere with the lactation process by affecting endocrine regulatory mechanisms. Study has pointed out that psychological distress may impair the release of oxytocin—a hormone that plays a key role in the milk ejection reflex during lactation, and its abnormal secretion can directly affect the normal excretion of breast milk (Nagel et al., 2022). When the lactation process becomes abnormal due to suppressed oxytocin secretion, it may not only lead to insufficient milk supply but also reduce the stability of breastfeeding. As an important way for infants and young children to obtain nutrition and establish emotional bonds, the decline in breastfeeding quality may further have potential adverse effects on the growth, development, and mental health of infants and young children. Chidiebele and other experts pointed out that inappropriate positioning of infants during breastfeeding may lead to imbalances in areas such as the neck, skull, and temporomandibular joint, thereby affecting the coordination and efficiency of sucking movements (Petronilla et al., 2018).

Therefore, Thus, studies indicate that in the absence of protective measures, breastfeeding activities may exert adverse effects on the health of both mothers and infants through multiple pathways, including musculoskeletal injuries and lactation abnormalities.

2.2. Comprehensive evaluation model

Evaluating whether new product development projects are timely and accurate significantly determines the efficiency of using scarce technological resources in a business, a sector, or even a country (Wei and Ruidan, 2001). To reduce the risk of unsuccessful new product launches, there are many evaluation methods used during the product development process. For example, Chen (2016) used questionnaire surveys, interviews, and material compilation methods to evaluate the product quality of e-commerce businesses. The Six Sigma DMAIC (Define, Measure, Analyze, Improve, Control) methodology can be adopted to assess and enhance the quality issues in the production process (Smętkowska and Mrugalska, 2018). These existing studies often involve both qualitative and quantitative issues, which cannot effectively handle some difficult-to-quantify, abstract, or vague subjective evaluations. However comprehensive evaluation methods can effectively solve this problem.

The comprehensive evaluation method has been widely used in numerous fields, and many studies on fuzzy comprehensive evaluation based on different theories exist. For instance, Qi et al. (2024) used grey system theory to determine indicator weights and established a multi-level fuzzy comprehensive evaluation model to assess the green construction of power grid projects. Chen et al. (2015) conducted



Fig. 1. Photos of some subjects during the preliminary experiment.

research on teaching performance evaluation based on the fuzzy analytic hierarchy process and fuzzy comprehensive evaluation theory. Xiaoyan et al. (2016) proposed a fuzzy comprehensive evaluation model for air traffic control safety risks based on the materi-element analysis theory. Yongjun and Xiaoyun (2012) based their fuzzy comprehensive evaluation model for the entrepreneurial environment of science parks on niche theory, among others.

In the field of ergonomic product design, the application of fuzzy comprehensive evaluation methods is also widely adopted. Incekara (2021) addressed health issues in children caused by poor sitting posture and inappropriate furniture design. Considering the impact of the COVID-19 pandemic, a fuzzy logic-based ergonomic classroom furniture design method was developed for primary school students aged 7 to 10 by measuring their body dimensions and applying fuzzy mathematical models, including fuzzy AHP, fuzzy TOPSIS, and fuzzy VIKOR. Yanhong et al. (2023) designed and evaluated household infant bathtubs using the fuzzy AHP. Starting from the user needs of household infant bathtubs, they established an evaluation model based on fuzzy AHP and conducted a fuzzy comprehensive evaluation of the design scheme, demonstrating the effectiveness of fuzzy AHP in product design evaluation for specific user groups. Qinglan and Yueqin (2024) used fuzzy AHP to construct a hierarchical indicator system and a fuzzy function evaluation matrix to quantify consumers' emotional needs. They analyzed the correlation between users' emotional intentions and various texture design elements, ultimately proposing an optimal combination of automotive leather seat textures that meet users' emotional preferences.

The comprehensive evaluation model has advantages such as multidimensional assessment, objectivity, scientific basis, decision support, and risk management. It can comprehensively evaluate matters from multiple aspects through quantitative indicators and weight settings, reducing the influence of human subjectivity. The model provides systematic analysis tools and consistent evaluation. It can quickly process large amounts of data, reveal hidden patterns and trends, and provide valuable insights, offering strong support for managers to make scientifically sound decisions.

Recent advancements in the field of engineering design highlight the importance of integrating user-centered approaches with technological innovation. However, existing research on nursing chair design reveals two main issues: a disconnect between technical feasibility and core user needs, and a lack of a quantitative evaluation framework that addresses both technological iteration and evolving user demands, particularly in key functions such as dynamic posture adaptation and pressure dispersion for relieving maternal fatigue.

To address these gaps, this study captures the technological evolution trends in nursing chair design through patent literature analysis and applies the Trapezoidal Fuzzy AHP to prioritize user requirement indicators. Based on this, a dynamic decision-making framework titled "Technology Trend Insight and Quantitative Indicator Evaluation" is constructed. This model anchors the boundaries of technical feasibility using patent data and quantifies core user needs through fuzzy weighting. Ultimately, it enables iterative optimization of design solutions, reduces the risk of product launch failure, and provides decision-making support that balances technical viability with user adaptability in nursing chair design.

3. Methodology

3.1. Demand-driven dynamic design decision framework

Breastfeeding poses significant challenges to the health of postpartum mothers, including musculoskeletal injuries caused by high frequency and prolonged nursing postures, as well as lactation abnormalities triggered by psychological stress through endocrine pathways. However, current nursing chair designs are still inadequate in meeting these multi dimensional needs, and the root cause lies in the failure to

effectively integrate engineering feasibility with users' subjective needs during the design process. Traditional product evaluation models often struggle to balance technical constraints and users' ambiguous perceptions, resulting in an imbalance among functionality, comfort, and user centricity.

To solve this problem, this study proposes a demand driven dynamic design decision making framework that integrates Patent Literature Analysis with the trapezoidal fuzzy Analytic Hierarchy Process (AHP) (see Fig. 2). Although both methods have been applied independently in previous studies, the methodological innovation of this framework lies in their systematic coupling, which improves the quality of decision making in design scenarios characterized by both technical complexity and user uncertainty. The methodological innovations are reflected in the following key aspects:

Coupled evaluation of technical feasibility and user demand prioritization: Unlike traditional fuzzy AHP models that rely entirely on subjective expert judgment to construct indicator systems and assign weights, this framework extracts high frequency keywords from Patent Literature Analysis to anchor fuzzy evaluation indicators within the scope of existing technical capabilities, thus ensuring their implementability from the very beginning. Subsequently, fuzzy AHP is used to quantify user priorities, forming a two way constraint mechanism - patent trends guide the construction of evaluation indicators, while fuzzy weights constrain the priority of technical implementation. This avoids unrealistic functional expectations and aligns design decisions with both user needs and technical reality.

This approach effectively bridges a long standing divide in design science: user centered design often overlooks technical feasibility, while technology driven design tends to ignore core user preferences. By combining Patent Literature Analysis with the trapezoidal fuzzy AHP, this framework builds a bridge between technical reality and vague user needs, forming an expandable and traceable decision making mechanism under uncertain conditions. It is particularly suitable for design optimization in complex product development scenarios.

The framework is implemented through three methodological modules: Patent Literature Analysis (PLA), Trapezoidal Fuzzy AHP, and fuzzy comprehensive evaluation, which are introduced in Sections 3.2 to 3.4.

3.2. Patent literature analysis (PLA)

Patent Literature Analysis is a major method of key technology analysis in the field of information science and is used to conduct technical and market research through patent literature (Matusun et al., 2024). By analyzing the trends in patent applications and grants in relevant fields, we can gain insights into the development status, research hotspots, and future trends of specific technological areas (Li et al., 2013). Unlike traditional literature reviews, PLA places greater emphasis on the traceability and foresight of technological innovations, thereby providing data-driven support for product design and technological development.

The process of PLA includes: (1) retrieval and screening of relevant patents; (2) extraction of high-frequency keywords and technological themes; (3) identification of the degree of association among high-frequency terms; (4) revelation of product development trends and inter-technology correlations. Through systematic analysis of this information, it becomes possible to better identify industry development trajectories, locate major innovation hotspots, and extract technological features related to product functionality.

In this study, PLA was applied to the retrieval and analysis of nursing chair-related patents in order to extract high-frequency keywords that reflect industrial trends and technological capabilities. These keywords not only serve as the basis for constructing the evaluation indicator system in the subsequent fuzzy Analytic Hierarchy Process (AHP), but also help ensure the feasibility and implementability of the indicators, thereby achieving a two-way coupling between user needs and

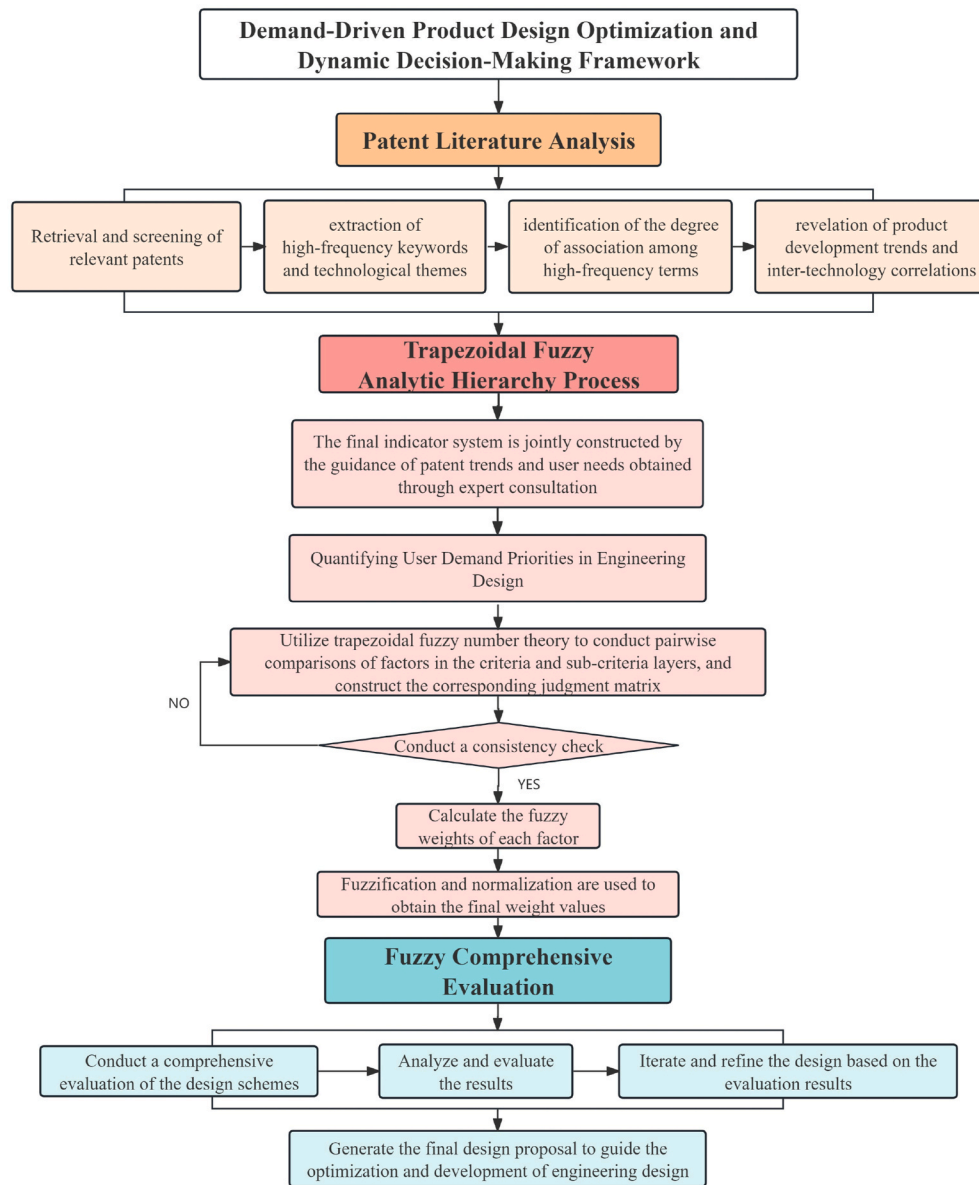


Fig. 2. Framework for demand-driven dynamic design decision-making.

engineering feasibility.

3.3. Trapezoidal fuzzy AHP

3.3.1. Trapezoidal fuzzy number

Although traditional AHP has systematic and structural advantages in analyzing demand weights (Liu et al., 2025), its reliance on precise numerical evaluations makes it difficult to handle the ambiguity and uncertainty in the decision-making process. In contrast, fuzzy AHP, by introducing fuzzy set theory, can more effectively capture and quantify these vague semantics (Chakraborty et al., 2017).

Trapezoidal fuzzy numbers were chosen over triangular or Gaussian membership functions for constructing the fuzzy AHP model due to their ability to more accurately represent the uncertainty and imprecision inherent in expert evaluations. Trapezoidal fuzzy numbers offer greater flexibility in capturing the range of possible values, which is particularly important in the context of nursing chair design evaluations where expert opinions can vary significantly. This flexibility allows for a more nuanced representation of uncertainty, leading to more robust and reliable decision-making. While triangular fuzzy numbers are simpler

and Gaussian functions provide smooth transitions, trapezoidal fuzzy numbers provide a balance between complexity and precision, making them a superior choice for this application.

A trapezoidal fuzzy number can be defined as $A = (a, b, c, d)$, where $a \leq b \leq c \leq d$; if $a > 0$, then A is called a positive trapezoidal fuzzy number.

In Equation (1), $A(x)$ is referred to as the trapezoidal fuzzy number, with a, b, c , and d being the membership function parameters of $A(x)$.

$$A(x) = \begin{cases} \frac{x-a}{b-a}, & a \leq x < b \\ 1, & b \leq x \leq c \\ \frac{d-x}{d-c}, & c < x \leq d \\ 0, & x < a, x > d \end{cases} \quad (1)$$

In this, a and d represent the upper and lower bounds of the support interval of the fuzzy set A , respectively, while the closed interval $[b, c]$ is the core of A . When $b = c$, it becomes a standard triangular fuzzy number.

3.3.2. Trapezoidal fuzzy AHP research flow

First, construct the evaluation criterion system of the trapezoidal fuzzy AHP, then invite experts, scholars, etc., to score the evaluation criteria and sub-criteria pairwise according to the 9-point relative importance scale (Table 1).

After obtaining the pairwise comparison results of the criteria and sub-criteria, establish the trapezoidal fuzzy evaluation matrix.

$$\tilde{A} = \begin{pmatrix} \tilde{a}_{11} & \tilde{a}_{12} & \cdots & \tilde{a}_{1n} \\ \tilde{a}_{21} & \tilde{a}_{22} & \cdots & \tilde{a}_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ \tilde{a}_{n1} & \tilde{a}_{n2} & \cdots & \tilde{a}_{nn} \end{pmatrix} \quad (2)$$

a_{ij} represents the attribute value of the i -th option on the j -th attribute, where each attribute value is expressed using a trapezoidal fuzzy number.

$$\tilde{a}_{ij} = (l_{ij}, m_{ij}, n_{ij}, s_{ij}) \quad (3)$$

$$\tilde{a}_{ji} = \tilde{a}_{ij}^{-1} = (s_{ij}^{-1}, n_{ij}^{-1}, m_{ij}^{-1}, l_{ij}^{-1}) \quad (4)$$

Secondly, in fuzzy multi-criteria decision-making problems, defuzzification is a key step. The Center of Gravity (COG) method was selected in this study because it can reflect the characteristics of fuzzy sets in a balanced manner and takes into account the overall distribution of the membership function. This method provides a mathematically concise and computationally efficient way to derive a representative crisp value, ensuring that the defuzzified output is both accurate and reliable (Feifei, 2020). While other defuzzification techniques such as the mean of maxima and bisector were considered, the COG method was selected due to its ability to handle the complexity and uncertainty inherent in the evaluation of nursing chair designs more effectively.

$X(A)$ is the defuzzified value, and for a trapezoidal fuzzy number $A = (a, b, c, d)$, the centroid method can be expressed in the form of equation (5).

$$X(A) = \frac{1}{3} \left[a + b + c + d - \frac{dc - ab}{(d + c) - (a + b)} \right] \quad (5)$$

Calculate the defuzzification of fuzzy numbers. The fuzzy evaluation matrix is converted into the following equation:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \quad (6)$$

Calculate the maximum eigenvalue of the matrix. The equation for calculating the maximum eigenvalue is as follows:

$$\lambda_{max} = \sum_{i=1}^n \frac{(AW)_i}{nW_i} \quad (7)$$

Table 1
9-point relative importance scale.

Scale of relative importance	Trapezoidal Fuzzy Number	Linguistic variable
1	(1,1,1,1)	Equally important
2	(3/5,7/4,9/4,5/2)	Slightly more important
3	(2,5/2,7/2,4)	Significantly more important
4	(7/2,15/4,17/4,9/2)	Moderately more important
5	(4,9/2,11/2,6)	Critically important
6	(11/2,23/4,25/4,13/2)	Moderately strongly more important
7	(6,13/2,15/2,8)	Very important
8	(15/2,31/4,33/4,17/2)	Moderately extremely more important
9	(8,17/2,19/2,10)	Absolutely important

Calculate the consistency index CI.

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (8)$$

Then calculate CR through CI.

$$CR = \frac{CI}{RI} \quad (9)$$

Here, the RI value is a random consistency index determined by the size of the judgment matrix. If the CR value is less than 0.1, the matrix is considered to have passed the consistency test; if the CR value is greater than 0.1, it indicates a problem with consistency (Yihan, 2024).

After the evaluation matrix passes the consistency test, apply the square root method (Equation (10)) to the trapezoidal fuzzy evaluation matrix to compute each factor, resulting in the fuzzy number for each factor, as follows:

$$r_i = (\tilde{a}_{i1} \times \tilde{a}_{i2} \times \cdots \times \tilde{a}_{in})^{-1} \quad (10)$$

a_{in} is the result of comparing the i -th factor with the n -th factor.

Then, calculate the fuzzy weight of the i -th factor according to Equation (11).

$$w_i = r_i \times (r_1 + r_2 + \cdots + r_n)^{-1} \quad (11)$$

At this time, $w_i = (w_{i1}, w_{i2}, w_{i3}, w_{i4})$.

For the fuzzy weights, first use Equation (5) for defuzzification, then normalize to obtain the normalized weight values for each factor.

Normalization formula:

$$W_i = \frac{w_i}{\sum_{i=1}^n w_i} \quad (12)$$

w_i represents the weight of the i -th factor after defuzzification, and W_i represents the normalized weight.

The traditional AHP is suitable for handling problems with deterministic data and high expert consensus (Miao et al., 2024). However, traditional AHP struggles to process fuzzy linguistic evaluations, whereas trapezoidal fuzzy AHP can effectively address problems with high uncertainty and significant divergence in expert opinions (e.g., innovative design evaluation, medical decision-making, etc.). In this study, the linguistic descriptions involved in Patent Literature Analysis and the inherent fuzziness in expert judgments regarding innovative designs make trapezoidal fuzzy AHP a more appropriate choice for this research.

The resulting weighted indicator system is further applied in a fuzzy comprehensive evaluation to generate final decision outputs, as detailed in Section 3.4.

3.4. Fuzzy comprehensive evaluation

Fuzzy comprehensive evaluation is a method based on the principles of fuzzy mathematics, used to handle comprehensive evaluations involving uncertain and ambiguous information (Jiafu et al., 2024). This method quantifies the impact of various factors on the evaluation object through fuzzy relational composition, leading to a comprehensive evaluation result. It is suitable for complex systems or decision-making problems that are difficult to describe with precise mathematical models.

The research process is as follows: According to the evaluation criteria parameters, this study adopted a 5-level evaluation system, use the evaluation levels V (Very Good (VG) = 0.9, Good (G) = 0.7, Medium (M) = 0.5, Poor (P) = 0.3, and Very Poor (VP) = 0.1) to evaluate the product. Each member of the expert panel must judge whether the product meets the evaluation criteria based on their knowledge and practical experience.

Construct a fuzzy evaluation matrix based on the evaluation indicators. Suppose there are m evaluation indicators in a certain level of

fuzzy evaluation, and the evaluation level is n . The evaluation data is a fuzzy matrix, expressed as follows:

$$R = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{pmatrix} \quad (13)$$

After obtaining the fuzzy evaluation matrix, the evaluation function is defined as shown in Equation (14), and the product's evaluation value can be calculated based on this formula.

$$E = W \times R \times V \quad (14)$$

Based on the results of Patent Literature Analysis and demand prioritization, this study generates high-potential design solutions. Using the fuzzy comprehensive evaluation method, the competitiveness of each design solution is assessed by combining expert scoring and weight values. The optimal solution is then selected for subsequent development based on the comprehensive evaluation scores.

4. Empirical example

4.1. Patent literature analysis

To demonstrate the practical applicability of the model, the study applied the proposed design solutions to a prototype nursing chair. The results show significant improvements in user comfort and ergonomic performance compared to existing designs. This suggests that our model can be effectively used to guide the development of other ergonomic furniture products.

To gain a deeper understanding of the research status and development trends in the field of product development, this study utilizes the ROST CM6 software to visualize high-frequency words and analyze the strength of associations between them in patent information. The patent data were extracted from the China National Knowledge Infrastructure (CNKI), a comprehensive academic database that provides access to a wide range of scholarly articles, theses, and patents. From a patent perspective, it explores research ideas, design hotspots, and development trends in the field of nursing chair design.

4.1.1. Visualization of high-frequency words

This study extracts selected patent abstracts from CNKI and processes them using the ROST CM6 software for word segmentation, filtering out unnecessary terms such as "invention," "utility model," and "design

patent." After segmentation, the high-frequency words are illustrated in Fig. 3.

4.1.2. Analysis of patent application hotspots

The patent data were obtained from China National Knowledge Infrastructure (CNKI), a comprehensive academic database that provides access to a wide range of scholarly articles, theses, and patents (Jie et al., 2022). This study retrieved the number of invention patents for nursing chairs filed in the CNKI database between January 1, 2013, and December 1, 2023. The selection of keywords and screening of patent documents follow the following criteria:

Keyword selection adheres to the core principle of "focusing on special seating for breastfeeding scenarios". It includes terms containing scenario-related words such as "breastfeeding" and "maternal-infant" along with form-related words such as "chair" and "seating" (e.g., "nursing chair", "breastfeeding chair"). It excludes terms for general-purpose seating (e.g., "children's dining chair") and non-seating maternal-infant products (e.g., "breastfeeding pillow").

Patent screening is conducted in three steps: first, a combined search of the selected keywords in titles and abstracts is performed, with duplicates removed using patent numbers; second, manual verification is carried out to exclude documents with irrelevant themes; third, only utility model patents and invention patents are retained (design patents are excluded as they have weak relevance to technological innovation trends).

The data show that the number of nursing chair patents in China has reached 93, with a growing trend in patent applications since 2016. A small peak in applications occurred in 2021. From a policy perspective, the National Health Commission and other departments jointly issued the Action Plan for Promoting Breastfeeding (2021–2025): China. 2021 (2024), which strongly promoted the construction of maternal and infant facilities, potentially stimulating innovations in nursing chairs to meet breastfeeding scenarios. In terms of population policy, the introduction of the "three - child policy" has attracted more investors to the maternal and infant industry, and nursing chairs, as a product related to maternal and infant care, have also been affected. In recent years, the number of patent applications for nursing chairs has declined, indicating that the technology in this field is maturing and awaiting the discovery of new technological growth points.

Using the ROST CM6 software, a co-occurrence matrix of words from the nursing chair patents was created, which includes the degree of association values between different vocabulary. The correlation heatmap was then calculated using RStudio software (Fig. 4), where red indicates a positive correlation, meaning both variables increase or decrease together; blue indicates a negative correlation, where one variable increases as the other decreases; and white or light colors indicate low correlation or no significant correlation (Haarman et al., 2015). The conclusions are as follows:

The correlation heatmap of patent word frequencies reveals a significant association between technological integration and functional diversity. The following steps summarize the trends based on the word-to-word correlations observed in the heatmap:

(1) Identify Strongly Correlated Word Clusters

Support Cluster: Support is strongly correlated with backrest, Armrest, Airbag, Nursing chair, and Both sides.

Structure-Material Cluster: Structure has strong correlations with Connection and Bracket, while Surface and Cushion form independent clusters.

Operation Cluster: Operation is strongly correlated with Convenient.

(2) Categorize Clusters by Functional Logic and Derive Trends

Trend 1: Adjustability and Multifunctional Integration.

Core Words: Adjustment, Height, Angle, Position, Device,



Fig. 3. Nursing chair high-frequency word tag cloud.

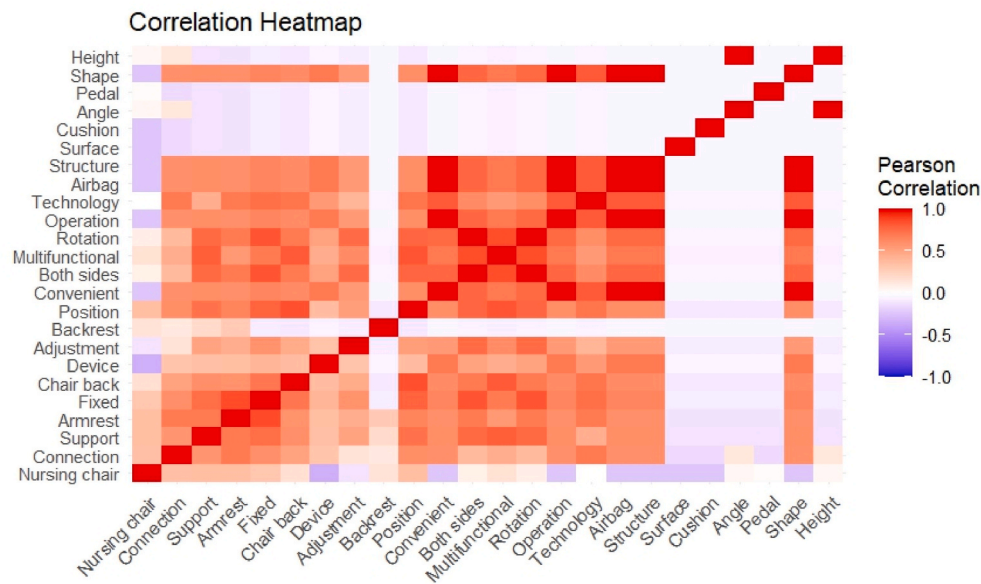


Fig. 4. Heatmap of high-frequency word correlations in nursing chair patent texts.

Multifunctional, Technology, Shape.

Correlation Logic: Adjustment is the "underlying technology": To adjust Height, Angle, and Position, it must rely on Device to achieve; Multifunctional is the "application scenario": With adjustment technology, the product can expand its functions; Technology is the "underlying support": Technologies such as sensors and intelligent control make the adjustment more accurate, and make the multifunctional more intelligent.

Trend Derivation: The multi-dimensional coverage of adjustment technologies, combined with the integration of multifunctional designs, points to the innovation direction of Adjustability and Multifunctional Integration.

Trend 2: Ergonomics and Dynamic Support

Core Words: Support, Chair back, Armrest, Airbag, Nursing chair, Both sides, Pedal.

Correlation Logic: Support is the "design core": The nursing chair must provide stable support for the human body, so Chair back and Armrest are the basis; Airbag is the "dynamic optimization": Ordinary support is "static", but the human posture will change, and the airbag can dynamically adjust the supporting force; Nursing chair is the "scenario-driven": The nursing scenario requires special support — for example, Both sides assists the patient to turn over, and Pedal adapts to the support needs during leg rehabilitation.

Trend Derivation: The coverage of "human adaptability" and "care-giving scenarios" by support technologies points to the innovation direction of Ergonomics and Dynamic Support.

Trend 3: Structural Optimization and Material Innovation.

Core Words: Structure, Connection, Bracket, Surface, Cushion.

Correlation Logic: Structure is the "macro carrier": Whether the nursing chair is stable and adjustable depends on Bracket and Connection; Surface and Cushion are the "micro experience": The parts that the user directly contacts need to be optimized independently. For example, Surface uses an antibacterial coating to prevent infection, and Cushion uses memory foam to improve comfort;

Trend Derivation: The "stability" "flexibility" of structural design, combined with the "functionalization" "adaptability" of material technologies, points to the innovation direction of Structural Optimization and Material Innovation.

Trend 4: Convenience and Intelligent Experience.

Core Words: Operation, Convenient.

Correlation Logic: Operation is "user interaction": Whether the nursing chair is easy to use depends on whether operations such as adjustment and switching are simple; Convenient is the "experience result": The simpler the operation, the more convenient the experience; **Implicit Correlation:** Convenient operation often relies on "intelligent technology", so it is indirectly related to Technology and Multifunctional.

Trend Derivation: The "convenience" of operational design, combined with the "intelligence" of technological implementation, points to the innovation direction of Convenience and Intelligent Experience.

(3) Strengthen Trends with Negative Correlation Verification

The strong negative correlation between Fixed and Adjustment reveals a conflict between fixed structures and adjustable functions, demonstrating that "breaking free from fixed designs and emphasizing adjustability" is an inevitable innovation direction (echoing the "Adjustability" trend).

If a negative correlation exists between Airbag and rotational functions, it indicates that dynamic support and rotational functions rarely coexist, suggesting the need to prioritize resolving core conflicts within the "Ergonomics and Dynamic Support" trend.

Through the classification of functional labels, analysis of strongly correlated word clusters, and verification of negative correlations for 24 high-frequency words in nursing chair patents, four major innovation trends are summarized: adjustability and multifunctional integration, ergonomics and dynamic support, structural optimization and material innovation, as well as convenience and intelligent experience. The analysis reveals the logic of technological synergy, contradictions to be resolved, and directions with innovation potential, providing a clear path from word correlations to design decisions for nursing chair research and development.

4.2. Trapezoidal fuzzy AHP

4.2.1. Building evaluation criteria

The evaluation criteria model for design requirements is a complex, multi-indicator, multi-level model where the selection of indicators must be comprehensive, independent, and cover all factors that could influence the evaluation results (Yinglong et al., 2022). To achieve this

objective, this study first conducted a Patent Literature Analysis (PLA) based on objective patent data related to nursing chairs, aiming to trace the logic of technological evolution and uncover potential correlations among technologies. Through word frequency analysis and the extraction of technological pathways, four major innovation trends in the nursing chair field were identified "Adjustability and Multifunctional Integration," "Ergonomics and Dynamic Support," "Structural Optimization and Material Innovation," and "Convenience and Intelligent Experience." These trends can further guide the construction of evaluation criteria.

However, to ensure that the evaluation model is more closely aligned with real-world usage scenarios, this study also incorporated the Delphi method. By gathering expert judgments on the actual usage needs of nursing chairs, the approach emphasizes both the rationality of user needs and the comprehensiveness of evaluation indicators. Experts proposed design requirements from the perspective of practical implementation, which, when combined with insights from patent trend analysis, enabled the construction of a final evaluation system that is both targeted and feasible.

As a result, the final evaluation framework consists of five primary criteria and twenty-one sub-criteria, as shown in Fig. 5.

The advantages of the two methods are complementary: Patent Literature Analysis (PLA) explores innovation possibilities and decomposes technological trends from the perspective of "technological history"; the Delphi method builds consensus and anchors the feasibility of evaluation from the perspective of "scenario needs". By combining the two methods, we ensure that the evaluation criteria are comprehensive, covering both technical aspects and user-centric aspects. These two methods jointly provide support for the research and development of nursing chairs and the sorting out of needs, and the design direction becomes clearer by integrating technological logic and consensus on needs.

4.2.2. Analyzing the priority of user requirements

A reasonable decision-making group is crucial for obtaining reliable evaluation results. To ensure knowledge breadth and reduce biases, our 30-member evaluation panel, consisting of 20 experts, 12 women with breastfeeding experience, and 8 maternal and infant furniture manufacturers, is formed in accordance with strict criteria:

The experts cover the fields of ergonomics and product design (16 persons) as well as maternal and infant care medicine (4 persons). The women with breastfeeding experience (aged 25–35) have recent breastfeeding experience, have used breastfeeding seating, and can

represent different demographic characteristics. The manufacturers have experience in the research, development or production of nursing chairs. Biases are reduced through multiple rounds of anonymous feedback: indicators with divergences are listed for discussion, and those with different opinions are required to provide basis with literature or cases until a consensus is reached.

In the evaluation of nursing chairs, indicators such as comfort are highly subjective. Trapezoidal fuzzy numbers allow evaluators to express their judgments through fuzzy intervals, capture uncertainties, and avoid biases caused by forced precise expressions, thus providing support for quantitative analysis. The evaluators scored each criterion using a 9-point scale (see Table 1), and after multiple rounds of expert discussions, the fuzzy judgment matrices for the criterion layer and sub-criterion layer were obtained (see Appendix A).

By comparing the criteria and sub-criteria pairwise, a trapezoidal fuzzy evaluation matrix was established. The trapezoidal fuzzy evaluation matrix was defuzzified using the centroid method (see Equation (5)), and its consistency was checked. Upon inspection, the consistency index of the data in this study was found to be less than 0.1, indicating good consistency of the matrix.

Subsequently, the fuzzy geometric mean of the trapezoidal fuzzy evaluation matrix was calculated (see Equation (8)). The fuzzy weight of each factor was obtained through Equation (9). After defuzzification (Equation (5)) and normalization (see Equation (10)), the final weight values were obtained as shown in Tables 2 and 3.

Through the trapezoidal fuzzy analytic hierarchy process for the standard and sub-standard layers of the nursing chair, a comprehensive

Table 2
Calculation results of criteria level weights.

Criteria level	Fuzzy weight vector	Defuzzified weights	Sort
A ₁ Functionality	(0.1690, 0.2174, 0.3545, 0.4688)	0.2838	2
A ₂ Comfort	(0.0895, 0.1118, 0.1733, 0.2213)	0.1396	3
A ₃ Aesthetics	(0.0339, 0.0429, 0.0655, 0.0819)	0.0524	5
A ₄ Convenience	(0.0339, 0.0420, 0.0684, 0.0941)	0.0561	4
A ₅ Scientific validity	(0.3012, 0.3856, 0.5887, 0.7276)	0.4682	1

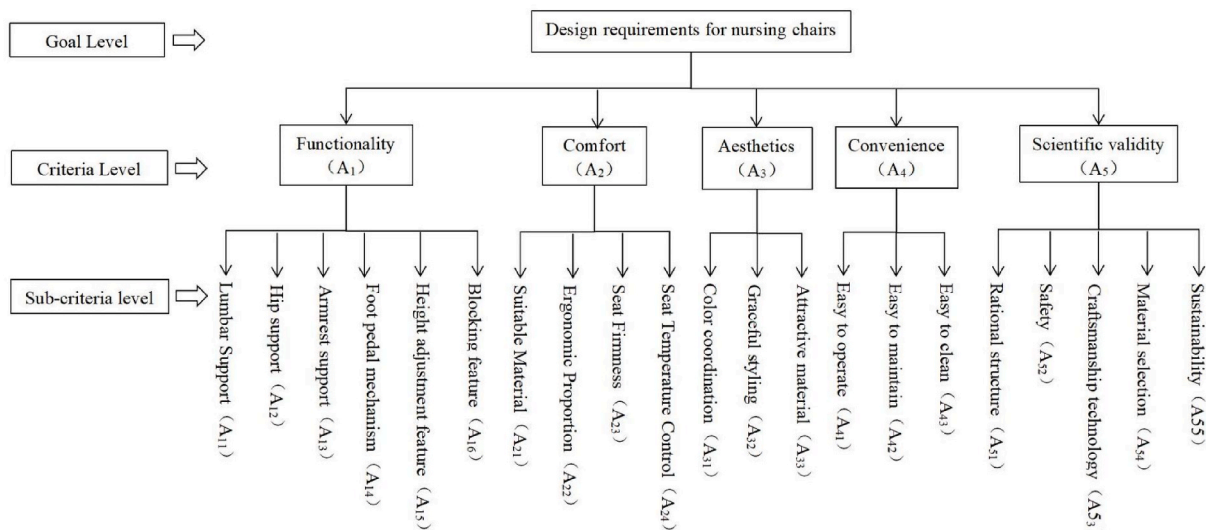


Fig. 5. Nursing chair design requirements analysis diagram.

Table 3
Calculation results of sub-criteria level weights.

Sub-criteria layer	Fuzzy weight vector	Defuzzified weights	Normalized weight	Sort
A ₁₁ Lumbar Support	(0.1661, 0.2069, 0.3155, 0.3931)	0.2561	0.0727	5
A ₁₂ Hip support	(0.0971, 0.1212, 0.1874, 0.2367)	0.1522	0.0432	7
A ₁₃ Armrest support	(0.2821, 0.3483, 0.5142, 0.6240)	0.4183	0.1187	3
A ₁₄ Foot pedal mechanism	(0.0572, 0.0710, 0.1098, 0.1393)	0.0894	0.0254	12
A ₁₅ Storage function	(0.0307, 0.0387, 0.0608, 0.0777)	0.0493	0.0140	15
A ₁₆ Blocking feature	(0.0228, 0.0274, 0.0416, 0.0539)	0.0346	0.0098	18
A ₂₁ Suitable material	(0.0659, 0.0839, 0.1300, 0.1638)	0.1067	0.0149	14
A ₂₂ Ergonomic Proportion	(0.4251, 0.5051, 0.6990, 0.8238)	0.5889	0.0822	4
A ₂₃ Seat Firmness	(0.1641, 0.1997, 0.2900, 0.3520)	0.2417	0.0337	10
A ₂₄ Seat Temperature Control	(0.0421, 0.0498, 0.0736, 0.0946)	0.0627	0.0088	19
A ₃₁ Color coordination	(0.1404, 0.1660, 0.2279, 0.2676)	0.1958	0.0103	17
A ₃₂ Graceful styling	(0.5495, 0.6292, 0.8162, 0.9301)	0.7136	0.0374	9
A ₃₃ Attractive material	(0.0667, 0.1034, 0.1242)	0.0906	0.0048	21
A ₄₁ Easy to operate	(0.5495, 0.6292, 0.8162, 0.9301)	0.7136	0.0400	8
A ₄₂ Easy to maintain	(0.0667, 0.0762, 0.1034, 0.1242)	0.0906	0.0051	20
A ₄₃ Easy to clean	(0.1404, 0.1660, 0.2279, 0.2676)	0.1958	0.0110	16
A ₅₁ Rational structure	(0.3213, 0.3930, 0.5688, 0.6825)	0.4689	0.2195	1
A ₅₂ Safety	(0.1802, 0.2212, 0.3295, 0.4065)	0.2717	0.1272	2
A ₅₃ Sustainability	(0.1027, 0.1245, 0.1855, 0.2312)	0.1539	0.0721	6
A ₅₄ Material selection	(0.0486, 0.0579, 0.0834, 0.1020)	0.0697	0.0326	11
A ₅₅ Craftsmanship technology	(0.0257, 0.0301, 0.0424, 0.0516)	0.0358	0.0168	13

model of fuzzy judgment for the nursing chair can be derived. In the evaluation of the standard layer indicators, scientific nature has the greatest impact on the design of the nursing chair, with a weight value of 0.4682, followed by functionality, comfort, aesthetics, and convenience. In the evaluation of the sub-criteria layer indicators, the weight values are ranked as follows: $A_{51} > A_{52} > A_{13} > A_{22} > A_{11} > A_{53} > A_{12} > A_{41} > A_{32} > A_{23} > A_{54} > A_{14} > A_{55} > A_{21} > A_{15} > A_{43} > A_{31} > A_{16} > A_{24} > A_{42} > A_{33}$.

4.3. Correlation analysis, validation, and design solution generation of model results

4.3.1. Comprehensive analysis of the correlation mapping between trends and indicators

The correlation mapping between the innovative trends derived from the patent analysis of nursing chairs and the indicators established through the trapezoidal fuzzy AHP is intended to bridge the gap between patent-based technological insights and specific design decisions. The trends identified through patent analysis reveal the macro-level direction of industry innovation, while the trapezoidal fuzzy AHP indicators quantify the priority weights of various design elements. This mapping not only transforms abstract innovation trends into actionable indicator systems but also integrates scattered indicators into coherent logical frameworks within each trend. It ultimately provides a clear and weighted foundation for decision-making in nursing chair design, enabling a precise alignment between technological innovation and user needs.

As shown in Table 4, each innovation trend is associated with a set of prioritized indicators based on their calculated weights:

Based on the overall weight distribution shown in Table 4, we propose the following categorization criteria to further clarify design priorities: High-priority indicators are defined as those with defuzzified weights greater than 0.10, representing core features that directly influence product performance, structural integrity, or user safety. Medium-priority indicators fall within the range of 0.05–0.10, referring to supportive elements that enhance ergonomic adaptability and user comfort. Low-priority indicators are those with weights below 0.05, typically reflecting auxiliary functions or aesthetic considerations, which should be optimized with minimal resource input. This classification provides a clear quantitative reference for determining the order of design investment.

The weighting system quantifies the importance of each indicator, thereby avoiding an equal distribution of design resources. High-weight indicators correspond to the core challenges within each trend and should receive focused investment. In contrast, low-weight indicators serve as supplementary functions that can be optimized without affecting the core design. This hierarchical design logic ensures the precise implementation of innovation trends, avoids unnecessary functional redundancy and cost waste, and ultimately achieves a balanced integration of technological feasibility and user needs.

4.3.2. Validation of model results based on industry standards and ISO norms

(1) Deep Alignment with Industry Ergonomic Design Standards

In the design of maternal and infant care products, GB/T 42803-2023 "Juvenile Products—General Requirements for Accessories of the Nursery Room" China, (2023) sets clear standards for nursing chairs in terms of structural stability (e.g., safety factor ≥ 1.5), adjustability (e.g., seat height suitable for users with a height of 150–180 cm), and comfort (e.g., lumbar support deviation ≤ 5 cm). The two indicators in this model with relatively high weights, namely "Structural Rationality" (weight: 0.2195) and "Lumbar Support" (weight: 0.0727), precisely align with the above requirements. This shows that the model's emphasis on core safety and ergonomic support is consistent with the established consensus in the industry.

The model assigns high weight to adjustable features such as Armrest Support (0.1187), which aligns with the "Adjustability and Multifunctional Integration" trend identified in patent analysis. This corresponds with the standard's requirement to accommodate the 95th percentile of user body dimensions, indicating that the model's logic for quantifying this trend effectively mirrors actual usage needs.

Furthermore, the model's treatment of auxiliary features demonstrates rational prioritization. For instance, Intelligent Temperature

Table 4
Mapping table of patent trends and design indicator weights for nursing chair.

Innovation Trend	Sub-criteria layer	Defuzzified weights	Sort	Mapping Rationale
Adjustability and Multifunctional Integration	A ₁₃ Armrest support	0.1187	3	Core to multi-dimensional adjustability, strongly supports ergonomic customization
	A ₁₄ Foot pedal mechanism	0.0254	12	Functional expansion requiring integration with adjustable components
	A ₁₅ Storage function	0.0140	15	Storage is a minor enhancement; needs to avoid interference with adjustment logic
	A ₁₆ Blocking feature	0.0098	18	Serves as a safety-enhancing add-on; should align with existing adjustable systems
Ergonomics and Dynamic Support	A ₁₁ Lumbar Support	0.0727	5	Essential to dynamic comfort and spinal alignment under prolonged use
	A ₁₂ Hip support	0.0432	7	Complements lumbar support for full lower-body ergonomic performance
	A ₂₂ Ergonomic Proportion	0.0822	4	Determines human-fit geometry, a foundation of ergonomic design
	A ₂₃ Seat Firmness	0.0337	10	Affects comfort over time but is easier to control than structure or shape
	A ₂₄ Seat Temperature Control	0.0088	19	Secondary comfort feature, with minimal impact on core ergonomic performance
	A ₂₁ Suitable material	0.0149	14	Material choice affects product adaptability and durability in usage environment
Structural Optimization and Material Innovation	A ₃₃ Attractive material	0.0048	21	Aesthetic concern with low functional influence; not a design priority
	A ₅₁ Rational structure	0.2195	1	Most critical factor for overall product integrity and innovation direction
	A ₅₄ Material selection	0.0326	11	Balances structural strength with tactile/aesthetic experience
	A ₅₅ Craftsmanship technology	0.0168	13	Supports feasibility and accuracy of

Table 4 (continued)

Innovation Trend	Sub-criteria layer	Defuzzified weights	Sort	Mapping Rationale
Convenience and Intelligent Experience	A ₅₃ Sustainability	0.0721	6	structure-related innovations Long-term strategy driver aligned with material use, recycling, and eco-innovation
	A ₃₁ Color coordination	0.0103	17	Visual harmony contributes to intuitive and calming user experience
	A ₃₂ Graceful styling	0.0374	9	Influences perception and acceptance of intelligent features
	A ₄₁ Easy to operate	0.0400	8	Central to user-friendly interfaces and smart controls
	A ₄₂ Easy to maintain	0.0051	20	Enhances overall experience but is not core to intelligent interaction
	A ₄₃ Easy to clean	0.0110	16	Important in healthcare environments, supports usability and hygiene
	A ₅₂ Safety	0.1272	2	Cross-functional imperative ensuring that convenience does not compromise safety

Control (0.0088) and Aesthetic Design are considered non-essential in the standard—and are similarly assigned lower weights in the model. This clear delineation ensures that design resources remain focused on core requirements while still addressing innovation trends like Dynamic Support Optimization, which are not yet covered by formal standards but frequently appear in patents.

Additionally, GB/T 18978.210–2024 "Ergonomics of Human-System Interaction" (2024) provides validation from a user experience perspective. It emphasizes that design should prioritize key user needs. The model reflects this by assigning a relatively high weight to Ease of Operation (A41, weight: 0.0400, ranked 8th overall), which aligns with ISO's focus on reducing user effort and complexity. This supports the technical feasibility and practical relevance of the indicator selection.

(2) Technical Validation Against ISO Standards

The model is further validated by its alignment with international benchmarks. For example, ISO 9241–5:2024 "Ergonomics of Human-System Interaction" [Standardization, \(2024\)](#) specifies that adjustable desks and seating should support the natural lumbar curve of the spine. This directly corresponds to the model's emphasis on the "Lumbar Support" indicator (A11, weight: 0.0727), which is calculated using the Trapezoidal Fuzzy AHP method. Patent analysis further supports this approach, as dynamic lumbar adjustment mechanisms frequently appear, bridging the gap between abstract ergonomic principles and implementable design features.

(3) Combined Value of Industry and ISO Standards

Validating the model through industry standards and ISO standards

enhances its credibility in three key dimensions: compliance, technical robustness, and user-centricity. Standards provide a solid baseline, while patent analysis offers forward-looking insights. By integrating these two aspects, it ensures that the model's design logic complies with regulatory requirements, is technically feasible, and can respond to user expectations, thus laying a reliable foundation for future innovations in nursing chairs.

4.3.3. Design solution generation

The rational design of ergonomic dimensions for nursing chairs has a positive impact on both mothers and infants. A nursing chair with improperly refined ergonomic dimensions for lactating women can potentially harm women's health.

The design of nursing chairs must be based on the seated human dimensions of women as the core basis, and all design schemes must strictly comply with ergonomic size specifications. According to the GB/T 10000-1988 "Human Dimensions of Chinese Adults" standard (2023), it provides detailed seated dimensions data for women aged 18–55 years. This data provides an important ergonomic reference for the design of nursing chairs. When applying dimension data, one should decide whether to use average values or specific percentile values based on practical needs (Longli, 2017). Regarding the research on the backrest tilt angle of nursing chairs, scholars like Santhosh and Gumasing have conducted studies (2019).

Based on the nursing chair design requirements derived from the analysis of this study, and under the premise of strictly adhering to ergonomic dimensions, two representative nursing chair design schemes have been developed, representing the two major design directions of "In-depth Development of Exclusive Scenarios" and "Integration with Family Scenarios" (Fig. 6). They differ significantly in terms of functional logic and application scenarios:

Proposal 1: exclusive design focused on lactation scenarios

This direction takes the "Multifunctional Modular Nursing Chair" as

Proposal 2: integrated design considering daily scenarios

This direction is represented by the "Intelligent Adjustment and Eco-friendly Nursing Chair", with the core of realizing natural integration into family scenarios through invisible function processing. In design, it weakens the tool attribute of "exclusively for lactation" and retains the appearance, form and basic functions of ordinary seats. This proposal adopts a simple Nordic style, which can be adapted to various home environments such as living rooms and studies. The manually adjustable armrests and head-neck rests are designed to not only meet the support needs during lactation but also serve as seats for daily office work and rest. Intelligent functions such as adaptive lumbar support exist in the form of "non-inductive adjustment", which neither affects the daily use experience of the chair nor provides customized support during lactation. This design is more in line with the needs of modern families for "no extra space occupation and no increase in home complexity", making the lactation function an "additional value" of ordinary seats rather than an independently existing "special furniture" (Fig. 6).

4.4. Fuzzy comprehensive evaluation

The two proposals were evaluated using the fuzzy comprehensive evaluation method. First, this study adopted a 5-level evaluation system, and 10 experts were invited to evaluate Proposal 1 based on the evaluation criteria parameters, with reference to the evaluation levels V (Very Good VG, Good G, Medium M, Poor P, Very Poor VP). Among them, $V = (0.9, 0.7, 0.5, 0.3, 0.1)$ was determined based on the core logic of the discrete fuzzy evaluation scale. It achieves a reasonable conversion from qualitative evaluation to quantitative values by establishing a one-to-one correspondence with the levels of the 5-level system. The summarized results are shown in Table 5.

Taking product functionality as an example, the fuzzy evaluation value is calculated as follows:

$$E_1 = \begin{pmatrix} 0.0727 & 0.0432 & 0.1187 & 0.0254 & 0.0140 & 0.0098 \end{pmatrix} \cdot \begin{bmatrix} 4 & 3 & 2 & 1 & 0 \\ 3 & 4 & 2 & 1 & 0 \\ 3 & 5 & 1 & 1 & 0 \\ 2 & 3 & 3 & 2 & 0 \\ 6 & 2 & 1 & 1 & 0 \\ 1 & 3 & 3 & 3 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0.9 \\ 0.7 \\ 0.6 \\ 0.3 \\ 0.1 \end{bmatrix} = 0.6847$$

the carrier, with the core of improving the lactation experience through scenario-specific optimization. In terms of design, it starts from the core needs of lactation behavior and deeply integrates functions such as height adjustment, supplies storage, and hygiene protection. This proposal features a one-touch height-adjustable seat, which can quickly adapt to the relative positions of the mother and baby; the storage area next to the armrest directly corresponds to items such as milk bottles and wet wipes that are frequently used during lactation; the antibacterial and waterproof fabric accurately solves the problem of cleaning up milk spills. Furthermore, through the modular design of the child feeding booster seat, the chair can be extended from the lactation period to the parenting period, realizing the exclusive value of "one chair for multiple uses" in specific scenarios. This proposal closely revolves around various needs of the lactation scenario, and comprehensively improves the convenience and comfort during lactation through precise matching and extension of functions.

Using the above equation, the fuzzy comprehensive evaluation values for the comfort, aesthetics, convenience, and scientific nature of the product in Proposal 1 can also be obtained, which are 0.7152, 0.6490, 0.6756, and 0.6426 respectively. The above values are presented in the order in which the indicators are listed.

The calculation of the comprehensive evaluation value for the nursing chair product in Proposal 1 is as follows:

$$E = (0.2838 \quad 0.1396 \quad 0.0524 \quad 0.0561 \quad 0.4682) \cdot \begin{bmatrix} 0.6847 \\ 0.7152 \\ 0.6490 \\ 0.6756 \\ 0.6306 \end{bmatrix} = 0.6669$$

Subsequently, Proposal 2 was evaluated following the identical procedure used for Proposal 1, with the same evaluation criteria and levels applied. The fuzzy comprehensive evaluation values for the functionality, comfort, aesthetics, convenience, and scientific nature of

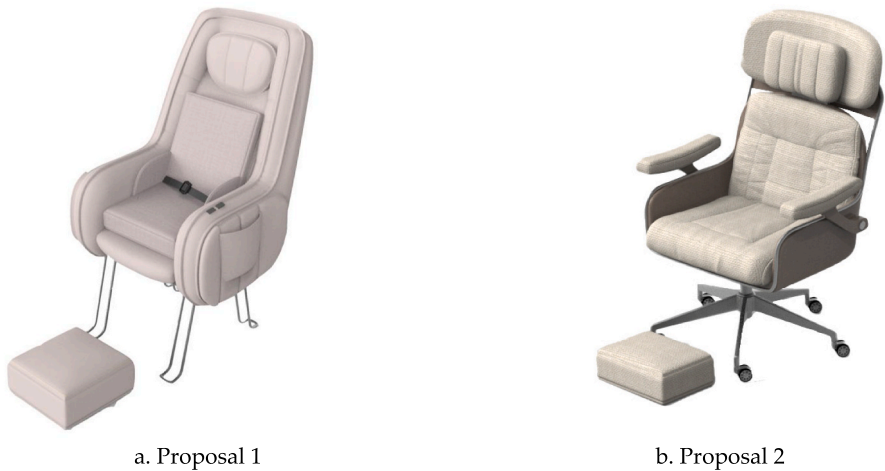


Fig. 6. Product proposal diagrams.

Table 5
Proposal 1 expert scoring sheet.

Sub-criteria	VG	G	M	P	VP
A ₁₁	4	3	2	1	0
A ₁₂	3	4	2	1	0
A ₁₃	3	5	1	1	0
A ₁₄	2	3	3	2	0
A ₁₅	6	2	1	1	0
A ₁₆	1	3	3	3	0
A ₂₁	4	4	1	1	0
A ₂₂	5	2	2	1	0
A ₂₃	4	3	2	1	0
A ₂₄	4	4	1	1	0
A ₃₁	2	3	3	2	0
A ₃₂	5	2	2	1	0
A ₃₃	2	2	4	2	0
A ₄₁	3	5	1	1	0
A ₄₂	1	2	6	1	0
A ₄₃	3	3	2	2	0
A ₅₁	4	3	2	1	0
A ₅₂	2	3	2	3	0
A ₅₃	3	5	1	1	0
A ₅₄	2	4	2	2	0
A ₅₅	2	4	3	1	0

Proposal 2 are 0.5965, 0.6107, 0.6181, 0.6709, and 0.6078, respectively. The final evaluation value of the product is 0.6092.

Based on the evaluation results, it is concluded that Proposal 1 is the superior solution, and subsequent improvement and development can proceed in accordance with this plan.

The proposed model integrating Patent Literature Analysis (PLA) and trapezoidal fuzzy AHP was benchmarked against other fuzzy MCDM techniques, including fuzzy TOPSIS and fuzzy DEMATEL, to demonstrate its comparative advantages. While fuzzy TOPSIS is effective in handling multiple criteria and providing a clear ranking of alternatives, it relies heavily on the concept of an ideal solution, which may not always be feasible in complex real-world applications. Fuzzy DEMATEL, on the other hand, excels in identifying and analyzing the interrelationships between criteria but can be computationally intensive and complex to implement. The proposed model, by integrating PLA and trapezoidal fuzzy AHP, offers a balanced approach that leverages the strengths of both methods. It provides a robust framework for handling uncertainty and subjectivity in expert evaluations while ensuring that the latest technological advancements are considered in the design

Table 6
Improvement suggestions for proposal 1 based on comprehensive evaluation.

Evaluation Criteria	Proposal 1 Score	Priority of User Demand Weight	Key Improvement Measures for Proposal 1 Based on Evaluation
Functionality	0.6847	Second highest (0.2838)	Partition the storage bag to enhance practicality; add classified compartments for different nursing supplies (such as feeding bottles and wet wipes) to improve access efficiency.
Comfort	0.7152	Third highest (0.1396)	When producing the product, optimize the density of the memory foam cushion to balance softness and support; add air vents in the seat fabric to enhance breathability during long-term use.
Aesthetics	0.6490	Fourth highest (0.0524)	Launch detachable and replaceable seat covers in various colors to match different home decoration styles; optimize the connection details of the aluminum alloy frame to reduce the visual sense of bulkiness.
Convenience	0.6756	Fifth highest (0.0561)	Design the seat height adjustment button as a large-sized convex type, facilitating blind operation by breastfeeding mothers.
Scientific Rationality	0.6426	Highest (0.4682)	1 Add safety belts to the infant feeding booster seat. 2 Reinforce the chair legs and install anti-slip rubber pads on them to enhance stability. 3 Add adaptive cushions based on pressure sensing to optimize the intelligent experience (Albert et al., 2024)

process. This combination not only enhances the accuracy and reliability of the decision-making process but also simplifies the computational complexity, making it more practical for real-world applications such as nursing chair design evaluations.

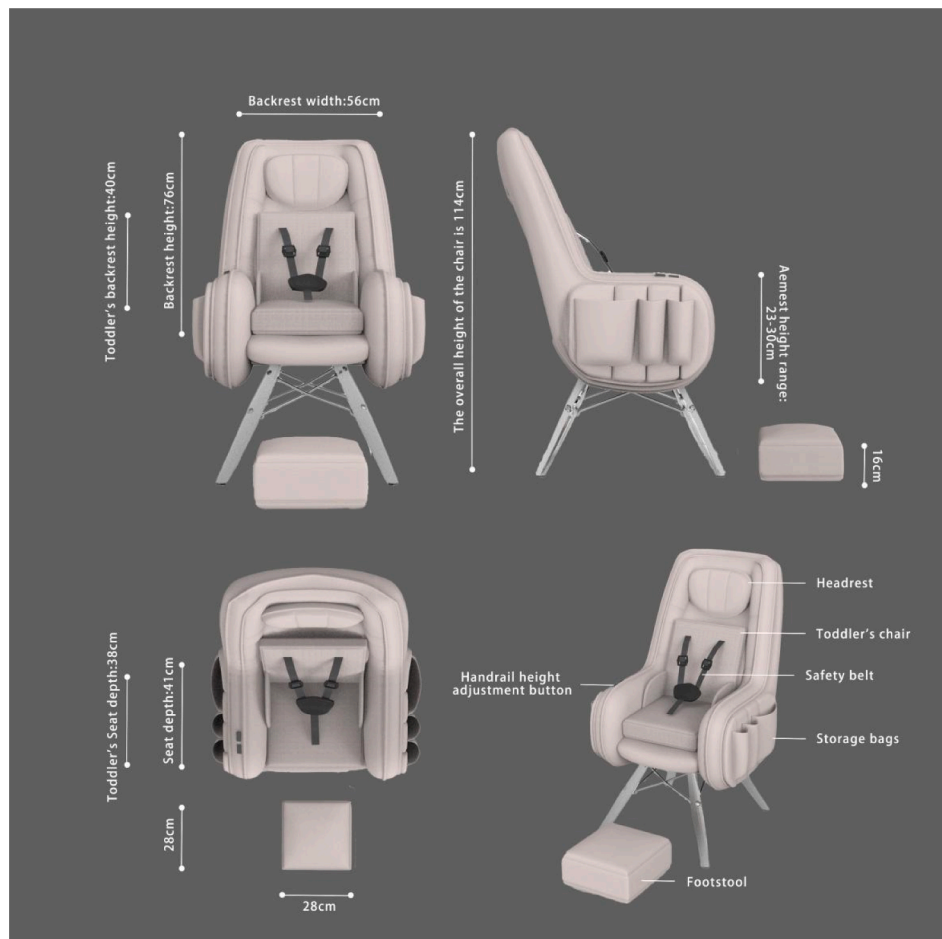


Fig. 7. Dimensional drawing of the new nursing chair design.

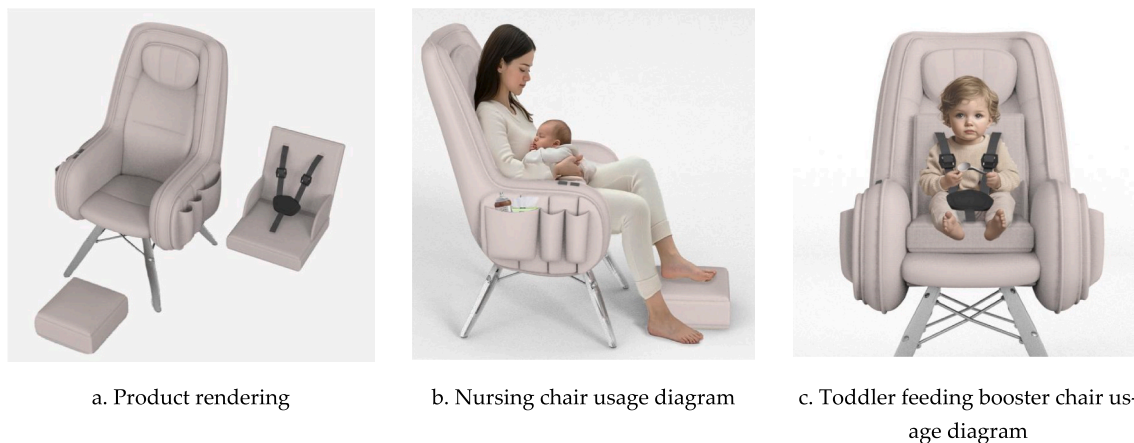


Fig. 8. Product solution of the new nursing chair design.

4.5. Proposal presentation

4.5.1. Comparison and optimization of proposals and user needs

Through in-depth studies of market and user demand evaluations, the direction of product design can be guided to ensure that it aligns with market trends and user expectations (Chunjin et al., 2024). Through the comprehensive evaluation of the nursing chair design, corresponding improvement suggestions for Proposal 1 have been put forward based on the comprehensive scoring results, as shown in Table 6.

4.5.2. Presentation of the improved proposals

Based on the comprehensive evaluation results summarized in the table, targeted improvement measures have been formulated for Proposal 1, covering adjustments in functionality, comfort, aesthetics, convenience, and scientific rationality in accordance with the priority of user demand weights. The revised design scheme of Proposal 1, incorporating all the above improvement suggestions, is presented in Figs. 7 and 8.

In summary, this study has identified the diverse needs and potential issues of nursing chairs and provided targeted solutions, aiming to create

a nursing chair that combines aesthetics, comfort, multifunctionality, and sustainability to meet the quality standards of modern households.

5. Discussion

This study identifies four major innovative trends in nursing chair design through patent literature analysis, and combines the trapezoidal fuzzy analytic hierarchy process to sort the weights of various indicators, ultimately forming a dynamic design decision-making framework. This combination model of "technological trend insight - quantitative indicator evaluation" shows significant advantages in complex product design decisions, which are specifically reflected in the following three aspects:

(1) Achieving precise coupling of technical feasibility and user needs

The core value of patent literature analysis lies in capturing the objective laws of technological evolution in the industry. The trends such as "adjustability integration" and "dynamic support optimization" extracted from nursing chair patents essentially reflect the feasibility boundaries at the technical level, that is, the functional boundaries that existing technologies can achieve. The trapezoidal fuzzy AHP, by quantifying the priority of user needs through fuzzy weights, clarifies the core demands that the design must prioritize to meet.

The combination of the two breaks the traditional dilemma of "disconnection between technological innovation and user needs": Patent trends provide "technical anchor points" for the trapezoidal fuzzy AHP indicator system, ensuring that all evaluation indicators are derived from practical and implementable technical solutions and avoiding "unrealistic demands"; and the weight ranking of the trapezoidal fuzzy AHP, in turn, restricts the priority of technical applications. For example, the "intelligent temperature control technology" that appears frequently in patents has a low weight in AHP (0.0088), so in design, it needs to be treated as a derivative function rather than a core investment, ensuring that resources are inclined to "structural safety" and "operational convenience" that users are more concerned about. This coupling mechanism enables design decisions to be rooted in technical reality and accurately respond to the pain points of needs. As Hagedorn et al. (2016) noted in medical device design research, traditional user-centered design often overlooks technical constraints, while technology-driven design tends to ignore need priorities, and integrating patent data with demand quantification can effectively resolve this contradiction.

(2) Enhancing the quantitative scientificity in fuzzy decision-making scenarios

Dynamic design decision-making often faces dual uncertainties: on one hand, users' evaluations of "comfort" and "convenience" are inherently ambiguous; on the other hand, the effectiveness of patented technologies fluctuates due to contextual influences. The trapezoidal fuzzy AHP addresses evaluation uncertainty by introducing fuzzy numbers. Upadhyay et al. (2021) demonstrated in their research on ergonomics in online education that employing fuzzy numbers to process subjective evaluations in fuzzy AHP can significantly reduce decision-making biases. Patent literature analysis, meanwhile, provides objective constraints for fuzzy weights.

For example, in the trend of "ergonomics and dynamic support," the fuzzy weight assignment for "lumbar support" does not rely solely on experts' subjective judgments but can be determined based on the high frequency of "lumbar adjustment structures" appearing in patents. This "fuzzy quantification + technical validation" model ensures that dynamic decision-making not only adapts to the uncertainties of complex scenarios but also maintains traceable scientific foundations.

(3) Strengthening the dynamic adaptability of design iteration

In the product life cycle, technological iteration and the evolution of user needs require design decisions to have the ability of dynamic adjustment. The combination of patent literature analysis and trapezoidal fuzzy AHP constructs a closed-loop mechanism of "trend update, weight iteration, and scheme optimization".

For example, when patent data shows an increase in the number of patent applications for "modular maintenance technology", it can trigger an update of the trapezoidal fuzzy analytic hierarchy process (AHP) indicator pool, incorporating "module compatibility" into the evaluation system for the "convenience" trend. Meanwhile, based on the technical maturity reflected in new patents, the trapezoidal fuzzy AHP can dynamically revise the weights of relevant indicators to promote the optimization of design schemes. This dynamic response capability transforms design decisions from static "one-time outputs" into a dynamic process that can continuously adapt to technological evolution and changes in demands. Bertheaux et al. (2019) also proposed a new product design method that integrates user-centered qualitative attributes. This method introduces mathematical formulas to process different types of data, converting users' qualitative preferences into quantitative design specifications. The formula is applied in the optimization process that considers the preference order of sensory attributes, and the solution to the optimization problem can provide the best user-centered specifications for the design concept of the final product. Therefore, in the aforementioned scenario of modular maintenance technology, this method can convert users' qualitative demands for module compatibility and ease of disassembly into specific quantitative indicators and integrate them into the design decision-making process. This enables design decisions to better meet user needs and maintain the optimality of multi-objective trade-offs when responding to technological changes and shifts in user demands, highlighting the importance of dynamic design decisions.

(4) Model Replicability and Cross-Product Generalizability

The design decision-making model that integrates patent literature analysis with the trapezoidal fuzzy analytic hierarchy process is not limited to nursing chair design; it demonstrates strong replicability and generalizability. Its core logic—identifying technological boundaries through patent analysis and quantifying user preferences via trapezoidal fuzzy AHP—can be directly applied to other complex product design contexts, such as baby strollers, elderly shower chairs, or ergonomic office chairs. The standardized workflow ensures that the model can be consistently implemented across various products, making it a universal tool for "technology–user demand synergy" in industrial design decision-making.

6. Conclusions

This study constructed a priority evaluation model for nursing chair design optimization through cross-analysis of Patent Literature Analysis and Trapezoidal Fuzzy AHP, and verified the feasibility of the model through preliminary tests.

From the perspective of user demand, comprehensive evaluation results of Proposal 1 showed clear rankings and key data across five core criteria.: comfort scored the highest (0.7152), followed by functionality (0.6847), convenience (0.6756), aesthetics (0.6490), and scientific validity, which scored the lowest (0.6426). It is worth noting that the article concludes through the analysis of patent technology trends and trapezoidal fuzzy AHP that scientific rationality has the highest weight in user demand, which indicates that the scientificity of the nursing chair in the design of Proposal 1 is one of the key priorities for improvement.

Aligning with both user demand priorities and the identified design optimization directions (high-priority: structural rationality (0.2195), safety (0.1272), etc.; medium/low-priority: ergonomic proportion, hip support, etc.), Proposal 1 was iteratively improved. Specifically,

targeting the low-scoring but high-weight scientific validity, improvements included adding safety belts to the infant feeding booster seat and enhancing chair leg stability—directly responding to the high-priority safety criterion (A52, 0.1272) and structural rationality (A51, 0.2195) identified in the fuzzy AHP-patent analysis. For functionality, which ranked second in user demand weight, the storage bag was partitioned to enhance practicality, aligning with the "adjustability and multifunctional integration" trend in patents. Additionally, adaptive cushions based on pressure sensing were added to optimize the intelligent experience, addressing both convenience (0.6756) and the "convenience and intelligent experience" patent trend, while also referencing research showing dynamic seating mitigates prolonged sitting effects (Albert et al., 2024).

These improvements verified that the design optimization path, based on multi-dimensional data fusion, effectively enhanced the scientific validity, functionality, and intelligent experience of nursing chairs. The research results provide a replicable methodological framework for the development of nursing chair products and even some office desks and chairs, helping to balance user needs, technological innovation, and engineering implementation. Beyond methodological contributions, this study also aligns with the United Nations Sustainable Development Goals. It supports Sustainable Development Goal 3, which aims to ensure healthy lives and promote well-being for all at all ages, by alleviating musculoskeletal strain and improving maternal and infant health outcomes. At the same time, it contributes to Sustainable Development Goal 9, which focuses on industry, innovation and infrastructure, by advancing innovative, data-driven approaches for product design and industrial upgrading. This highlights the dual academic and societal value of the proposed framework, bridging ergonomic research, industrial innovation, and global sustainability agendas (United Nations, 2023).

7. Limitations

(1) Limitations in Data Samples and Sources

This model relies on expert evaluations and patent literature analysis, but the quantity and sources of samples are limited, which may affect the generalizability of the conclusions. Although patent data can reflect industry trends, it is difficult to fully cover cutting-edge innovations and is often influenced by regional or corporate strategic factors. Constrained by the privacy of breastfeeding behavior, large-scale user surveys are challenging to conduct, resulting in limited numbers and diversity of participants. Specific groups such as women who have undergone cesarean sections, mothers of multiples, and users with disabilities are not systematically represented.

To address this limitation, future research will incorporate direct user testing and ergonomic trials involving postpartum women to validate expert-based assessment results. This improvement will ensure that design schemes not only meet expert standards but also satisfy the actual needs of end-users, thereby providing more comprehensive validation for the proposed designs.

(2) Potential Biases in the Evaluation Process.

Although the research team has nursing experience and has attempted to balance gender representation, there remains an underrepresentation of female experts and limited direct user participation. This may introduce subjectivity in the weighting of user needs. Furthermore, in the trapezoidal fuzzy AHP, the assignment of fuzzy numbers relies heavily on expert experience. Without robust verification mechanisms, these subjective inputs may amplify bias and compromise the accuracy of the final decision-making outcomes.

(3) Contextual Limitations of Model Applicability

The model is most effective for product designs with clearly defined technological trends and quantifiable indicators. In early-stage innovation contexts or areas lacking mature patent data, its feasibility and effectiveness are constrained. Additionally, the model primarily focuses on static functions and ergonomic compatibility, with limited consideration for soft-value factors such as emotional resonance or cultural adaptability—limiting its application in user experience-driven product design.

(4) High Resource Dependence for Dynamic Adjustment

Although the model constructs a closed-loop mechanism of "trend updates – weight adjustments – solution optimization," its operation relies heavily on ongoing patent monitoring, expert collaboration, and real-time data feedback. In scenarios with insufficient resources or delayed data updates, it becomes difficult to ensure the model's dynamic responsiveness and timely decision-making effectiveness.

8. Future research

(1) Expansion of Samples and Data Dimensions

Increase sample size in user surveys by adopting anonymization and privacy protection technologies to improve participation rates.

Combine quantitative questionnaires with qualitative observations (e.g., behavior tracking of lactating mothers) to comprehensively assess diverse needs.

Incorporate a wider variety of nursing chair products (such as high-end customized models and rehabilitation-oriented designs) along with cross-disciplinary data to enhance model robustness.

(2) In-Depth Optimization of User Participation

Introduce user testing and focus groups to gather direct feedback on usage experiences and pain points from lactating mothers, integrating this first-hand data into design iterations.

Establish a tripartite collaborative evaluation system involving experts, users, and technology to improve the accuracy and inclusiveness of demand weight assessment.

(3) Methodological Innovation and Application Extension

Explore the integration of machine learning algorithms with trapezoidal fuzzy AHP to dynamically optimize the design priority model.

Extend the research framework to maternal and infant care products (e.g., nursing pillows, baby cribs) to develop a systematic design theory for healthcare products.

CRedit authorship contribution statement

Jun Du: Writing – original draft. **Jiangang Zhu:** Writing – review & editing.

Data

Data will be made available on request.

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Declaration of competing interest

No potential conflict of interest was reported by the authors.

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Appendix A

Tables A1–A6 in this section show the process of pair-wise comparison analysis of criteria

Table A1

Pair-wise comparison matrix of the criteria.

A	A ₁	A ₂	A ₃	A ₄
A ₁	(1,1,1,1)	(2,5/2,7/2,4)	(6,13/2,15/2,8)	(2,5/2,7/2,4)
A ₂	(1/4,2/7,2/5,1/2)	(1,1,1,1)	(4,9/2,11/2,6)	(2,5/2,7/2,4)
A ₃	(1/8,2/15,2/13,1/6)	(1/6,2/11,2/9,1/4)	(1,1,1,1)	(1,3/2,5/2,3)
A ₄	(1/4,2/7,2/5,1/2)	(1/4,2/7,2/5,1/2)	(1/3,2/5,2/3,1)	(1,1,1,1)
A ₅	(1,3/2,5/2,3)	(4,9/2,11/2,6)	(6,13/2,15/2,8)	(6,13/2,15/2,8)

Table A2

Pair-wise comparison matrix of the sub-criteria.:A1 Functionality.

A ₁	A ₁₁	A ₁₂	A ₁₃	A ₁₄	A ₁₅	A ₁₆
A ₁₁	(1,1,1,1)	(2,5/2,7/2,4)	(1/4,2/7,2/5,1/2)	(3,7/2,9/2,5)	(4,9/2,11/2,6)	(5,11/2,13/2,7)
A ₁₂	(1/4,2/7,2/5,1/2)	(1,1,1,1)	(1/5,2/9,2/7,1/3)	(2,5/2,7/2,4)	(3,7/2,9/2,5)	(4,9/2,11/2,6)
A ₁₃	(2,5/2,7/2,4)	(3,7/2,9/2,5)	(1,1,1,1)	(4,9/2,11/2,6)	(5,11/2,13/2,7)	(6,13/2,15/2,8)
A ₁₄	(1/5,2/9,2/7,1/3)	(1/4,2/7,2/5,1/2)	(1/6,2/11,2/9,1/4)	(1,1,1,1)	(2,5/2,7/2,4)	(3,7/2,9/2,5)
A ₁₅	(1/6,2/11,2/9,1/4)	(1/5,2/9,2/7,1/3)	(1/7,2/13,2/11,1/5)	(1/4,2/7,2/5,1/2)	(1,1,1,1)	(1,3/2,5/2,3)
A ₁₆	(1/7,2/13,2/11,1/5)	(1/6,2/11,2/9,1/4)	(1/8,2/15,2/13,1/6)	(1/5,2/9,2/7,1/3)	(1/3,2/5,2/3,1)	(1,1,1,1)

Table A3

Pair-wise comparison matrix of the sub-criteria.:A₂ Comfort.

A ₂	A ₂₁	A ₂₂	A ₂₃	A ₂₄
A ₂₁	(1,1,1,1)	(1/6,2/11,2/9,1/4)	(1/4,2/7,2/5,1/2)	(1,3/2,5/2,3)
A ₂₂	(4,9/2,11/2,6)	(1,1,1,1)	(3,7/2,9/2,5)	(6,13/2,15/2,8)
A ₂₃	(2,5/2,7/2,4)	(1/5,2/9,2/7,1/3)	(1,1,1,1)	(4,9/2,11/2,6)
A ₂₄	(1/3,2/5,2/3,1)	(1/8,2/15,2/13,1/6)	(1/6,2/11,2/9,1/4)	(1,1,1,1)

Table A4

Pair-wise comparison matrix of the sub-criteria.:A₃ Aesthetics.

A ₃	A ₃₁	A ₃₂	A ₃₃
A ₃₁	(1,1,1,1)	(1/6,2/11,2/9,1/4)	(2,5/2,7/2,4)
A ₃₂	(4,9/2,11/2,6)	(1,1,1,1)	(5,11/2,13/2,7)
A ₃₃	(1/4,2/7,2/5,1/2)	(1/7,2/13,2/11,1/5)	(1,1,1,1)

Table A5

Pair-wise comparison matrix of the sub-criteria.:A₄ Convenience.

A ₄	A ₄₁	A ₄₂	A ₄₃
A ₄₁	(1,1,1,1)	(5,11/2,13/2,7)	(4,9/2,11/2,6)
A ₄₂	(1/7,2/13,2/11,1/5)	(1,1,1,1)	(1/4,2/7,2/5,1/2)
A ₄₃	(1/6,2/11,2/9,1/4)	(2,5/2,7/2,4)	(1,1,1,1)

Table A6
Pair-wise comparison matrix of the sub-criteria: A₅ Scientific validity.

A ₅	A ₅₁	A ₅₂	A ₅₃	A ₅₄	A ₅₅
A ₅₁	(1,1,1,1)	(2,5/2,7/2,4)	(3,7/2,9/2,5)	(5,11/2,13/2,7)	(6,13/2,15/2,8)
A ₅₂	(1/4,2/7,2/5,1/2)	(1,1,1,1)	(2,5/2,7/2,4)	(4,9/2,11/2,6)	(5,11/2,13/2,7)
A ₅₃	(1/5,2/9,2/7,1/3)	(1/4,2/7,2/5,1/2)	(1,1,1,1)	(3,7/2,9/2,5)	(4,9/2,11/2,6)
A ₅₄	(1/7,2/13,2/11,1/5)	(1/6,2/11,2/9,1/4)	(1/5,2/9,2/7,1/3)	(1,1,1,1)	(3,7/2,9/2,5)
A ₅₅	(1/8,2/15,2/13,1/6)	(1/7,2/13,2/11,1/5)	(1/6,2/11,2/9,1/4)	(1/5,2/9,2/7,1/3)	(1,1,1,1)

Appendix B

Proposal 2 Expert Scoring Sheet

Sub-criteria	VG	G	M	P	VP
A ₁₁	2	3	4	1	0
A ₁₂	1	2	2	5	0
A ₁₃	3	3	2	2	0
A ₁₄	2	4	2	2	0
A ₁₅	1	2	4	3	0
A ₁₆	1	2	1	6	0
A ₂₁	3	3	3	1	0
A ₂₂	2	2	4	2	0
A ₂₃	2	4	3	1	0
A ₂₄	4	3	2	1	0
A ₃₁	3	2	3	2	0
A ₃₂	2	4	2	2	0
A ₃₃	2	3	3	2	0
A ₄₁	3	4	2	1	0
A ₄₂	1	3	5	1	0
A ₄₃	4	2	3	1	0
A ₅₁	2	3	4	1	0
A ₅₂	2	2	4	2	0
A ₅₃	1	3	3	3	0
A ₅₄	2	3	4	1	0
A ₅₅	1	3	4	2	0

Data availability

I have shared the link to my data at the research paper

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